A Case Study of Environmental Footprints for Generative AI Inference: Cloud versus Edge

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ABSTRACT

The rapid growth of generative AI has placed significant strain on traditional data center infrastructures and existing power grids, leading to soaring energy demands and environmental burdens that may disproportionately affect the local communities. Shifting AI inference from the cloud to edge devices could potentially reduce the reliance on network connections, enhance user privacy, and alleviate the escalating pressure data centers impose on the local electricity grid. In this work, we present a case study examining the environmental footprint and energy consumption when deploying a generative AI model on cloud and edge platforms. To this end, we model and evaluate the water consumption and carbon emissions associated with AI inference across these deployment scenarios. Our empirical results demonstrate that, for several state-of-the-art generative AI models deployable on both cloud and edge devices, a reduced environmental footprint is observed for edge platform deployments. More specifically, edge platforms can achieve over 90% energy savings while reducing carbon emissions and water consumption by more than 80%. Putting the accuracy and latency performance aside, these findings highlight the potential of edge inference to lower the energy demands and environmental footprint of generative AI compared to cloud-based inference.

1. INTRODUCTION

The proliferation of generative AI has accelerated significantly in recent years, driven by the immense potential across diverse fields, including healthcare, education, drug discovery, and climate modeling. Many generative AI models, particularly those with expansive parameter sets, are typically trained and deployed in energy-intensive data centers. In 2022, data centers consumed roughly 2% of global electricity [1], contributing approximately 1% of global greenhouse gas emissions [2]. Furthermore, the AI market and energy consumption is projected to expand tenfold within the next five years , driven by the escalating demand for generative AI. This surge will pose significant challenges to existing data center infrastructure and energy systems. Not only the physical space hinders data center capacity expansion, limitations imposed by power capacity are also particularly acute, especially in regions with large data center concentrations such as Virginia. Industry leaders have mirrored this trend by calling for an imperative breakthrough in power systems to meet the immense energy requirements of the burgeoning AI industry [3].

Simultaneously, the expanding processing capabilities of edge devices have recently spurred a potential shift in computational paradigms, transitioning from centralized data centers to decentralized edge devices such as smartphones. Offloading the AI inference from the cloud to edge devices helps to minimize the effects of unreliable network connections [4], which is essential to improve the users' quality of experience. Moreover, by storing both the model weights and users inputs locally, the on-device inference can better safeguard user privacy and avoid the leakage of sensitive information.

In addition, the energy consumption of edge computing is more distributed than cloud computing, as the cloud computation platforms are typically hosted in warehouse-level data centers. Considering the escalating demand of AI computing, such concentrated energy pressure of datacenter would not only drain the local electricity capacity, but could also lead to disproportionate social and environmental costs [5]. For instance, the concentrated electricity consumption required to power data centers in Virginia can contribute to higher transmission congestion costs, carbon emissions and/or hazardous solid wastes. Additionally, water evaporation needed for electricity generation and data center cooling [6] can exacerbate freshwater scarcity in certain regions.

Quantifying the environmental footprint of such AI model inferences is a non-trivial task, requiring distinct considerations for both cloud and edge platforms. For carbon emissions, we not only account for emissions generated during electricity production but also utilize life cycle analysis (LCA) tools to evaluate embodied carbon. A significant difference exists between the carbon emissions associated with manufacturing and the nominal lifespans of cloud servers versus mobile devices, resulting in varying embodied carbon impacts for cloud and edge platforms. Similarly, water consumption during AI inference arises partly from electricity generation on both cloud and edge platforms. However, cloud platforms also contribute to additional water consumption through onsite evaporation[7] required for server cooling.

That being said, leveraging the centralized infrastructure, cloud platforms excel at scalability and parallel data processing. Besides, these powerful cloud servers can handle extensive computational workloads and support most advanced AI models. On the other hand, edge platforms, which deploy AI models closer to the data source—such as on

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mobile devices, IoT devices, or local servers—often attain better energy efficiency. By minimizing the need for data transmission to distant cloud servers, edge computing also reduces latency and energy consumption, thereby lowering the environmental impact of generative AI inference.

Recent studies have demonstrated that edge platforms could be more environmentally sustainable than cloud-based solutions, particularly when running some machine learning (ML) models. For instance, research [8, 9] highlights the substantial reduction in carbon footprint achieved by deploying AI on edge devices. These findings highlight the potential of edge computing to mitigate the environmental costs associated with AI, especially as the demand for generative AI continues to grow. By leveraging localized processing, edge platforms not only enhance energy efficiency but also contribute to a more sustainable AI ecosystem, aligning with global efforts to reduce carbon emissions and promote greener technologies.

Our empirical findings provide additional insights by considering a concrete case study in environmental footprint evaluation between cloud and edge platforms, which includes both water consumption and carbon emissions. Our methodology helps understand the environmental benefits of edge inference over their cloud-based alternatives. Specifically, we compare the energy usage and environmental footprint of several popular up-to-date generative models running on both cloud servers with high-end GPUs (e.g., Nvidia A100) and edge devices (e.g., Samsung S24). Using identical model inputs and architectures, our results suggest that edge inference can potentially achieve a 90% energy saving and an 80% environmental footprint reduction compared to cloud inference for generative AI.

2. METHODOLOGY OVERVIEW

We evaluate the inference time, energy consumption, and environmental footprints of several popular models on both cloud and edge devices. In our main experiment, we compare these metrics for several generative AI models deployed on cloud servers with either Nvidia A100 or L4 GPUs (hosted on Google Colab) and the Samsung S24 edge device.

Profiling Metrics: In order to estimate the environmental footprints, we profile generative AI model performance with a focus on two key metrics: inference time and energy consumption. The inference time, or end-to-end latency, is the elapsed time between a user submitting a request and receiving the generative model's response, excluding model loading and data preprocessing. Similarly, energy consumption is measured during this inference process. The software kits for measuring inference and energy consumption on cloud and edge are detailed in Section 3.2.

Environmental Footprint: Based on the measurements, we estimate the environmental footprint, including carbon footprint and water consumption incurred by generative AI model inference. Apart from the operational carbon emission during the model inference, we further incorporate the embodied missions during manufacturing, transport, and recycling. Our analysis is based on the widely used Life Cycle Analysis (LCA) tools, following the practices of many large companies. More specifically, the carbon footprint of deploying a generative model i on device d is given by

$$C_{(i,d)} = \gamma_c \cdot E_{(i,d)} + \frac{O_d}{U_d \cdot L_d} \cdot T_{(i,d)} \quad (1)$$

where $E_{(i,d)}$ and $T_{(i,d)}$ are the energy consumption (including applicable overheads such as data center cooling and mobile phone charging loss) and inference time associated with model i on device d, γ_c denotes the average carbon intensity of electricity generation, and O_d, L_d, and U_d denote the life-cycle carbon emission, the nominal lifetime and utilization rate of device d, respectively. The average carbon intensity γ_c is determined by the average regional fuel mix of electricity generation. The two terms in Eqn. (1) denotes the operational carbon emission and amortized embodied carbon during the inference period, respectively

In the cloud computing environment, the energy consumption, denoted as $E_{(i,d)}$, includes not only the energy used by IT equipment for computational tasks, but also the energy consumed by non-IT infrastructure, such as cooling systems, power distribution units, and other supporting facilities. This comprehensive energy consumption is typically calculated by incorporating the Power Usage Effectiveness (PUE), represented as δ_d, which serves as a multiplier to account for the energy overhead associated with non-IT equipment. The PUE metric is a critical factor in evaluating the energy efficiency of data centers, as it quantifies the ratio of total facility energy to the energy consumed solely by IT devices. A lower PUE value indicates a more energy-efficient data center, as less energy is wasted on non-computational activities.

On the other hand, when considering edge devices, such as smartphones, IoT devices, or local servers, the energy consumption model also account for additional factors beyond just the operational energy for a consistent comparison. One significant aspect is the energy overhead incurred during the battery charging process. For instance, energy losses occur during the conversion of alternating current (AC) from the power grid to direct current (DC) required by the device's battery. These losses, often referred to as charging inefficiencies, can substantially impact the overall energy footprint of edge devices. Additionally, factors such as battery degradation over time, thermal management, and idle power consumption further contribute to the total energy expenditure.

By incorporating these considerations into the energy consumption models for both cloud and edge environments, researchers and practitioners can gain a more accurate understanding of the energy dynamics involved in deploying AI and other computational workloads. This holistic approach enables the identification of opportunities for optimizing energy efficiency, reducing environmental impact, and making informed decisions about the trade-offs between cloud and edge computing platforms.

Due to the lack of reliable and comprehensive data sources for conducting a life cycle analysis (LCA) of embodied water consumption across the supply chain, this study narrows the the water footprint assessment of both cloud and edge platforms to operational water consumption. Operational water consumption primarily refers to the water utilized during the active use phase of AI systems, such as the cooling of data centers and the operation of servers, which are critical for maintaining optimal performance and preventing overheating in both cloud and edge computing environments. To sum up, the methodology of the operational water footprint is similar to the operational carbon footprint, by simply replacing the carbon intensity γ_c with water usage efficiency γ_w.

3. EXPERIMENT SETUP

This section first describes the experimental platform and profiling tools used in our empirical study. Subsequently, we detail the model and dataset employed for generative model inference. Finally, we present the latency profiling methodology for generative AI models on cloud and edge platforms.

3.1 Experiment Platform

We choose Google Colab as the cloud environment, which offers affordable access to a range of server configurations, spanning from servers equipped with NVIDIA's A100 GPUs to those featuring the highly efficient L4 GPUs. Specifically, our profiling analysis on Google Colab utilizes two distinct configurations: the first features one NVIDIA A100 GPU (with 40 GB GPU RAM), 83.5 GB system RAM, and 201.2 GB allocated disk space, while the second employs one NVIDIA L4 GPU (with 22.5 GB GPU RAM), 53.0 GB system RAM, and 201.2 GB allocated disk space. This setup allows us to compare performance across different GPU architectures within the Colab environment.

For the on-device analysis, we select devices based on their capacity to deploy generative models, considering the resource constraints of edge devices. Concretely, our primary experiment is conducted with Samsung Galaxy S24, equipped with the Qualcomm Snapdragon 8 Gen 3 chipset. The chipset is the most advanced Snapdragon mobile platform at the time of these experiments. This configuration ensures compatibility with a diverse set of generative models. The inference time of the generative models are mainly profiled using the Neural Processing Unit (NPU) within the Qualcomm chipset, which is specially designed for accelerating the AI workloads. The measurement of NPU-based inference is implemented on the cloud-hosted edge device platform, Qualcomm AI hub, supporting the deployment of generative models on NPUs. To provide a more comprehensive comparison, we also benchmark the performance of on-device AI model deployment on GPUs of several mobile devices, including the devices (e.g., Google Pixel 8). Google Pixel 8, released in the same year as the Samsung Galaxy S24, provides a set of additional data points for our analysis. The detailed profiling of model performance on the Google Pixel 8 can be found in our full technical report [10], further validating our comparison of environmental footprints between cloud and edge devices.

3.2 Profiling Tools

Compared with edge devices, cloud servers often have better tools and support to deploy generative AI models and measure their latency and energy in an end-to-end manner. Unless specified otherwise, all the models profiled on the cloud are downloaded from Huggingface and executed with their official transformer API. The inference time and consumed energy for each prompt is collected using the context manager provided by CodeCarbon [11], which records the elapsed time solely for the model inference code segment and measures the energy usage of the server components (e.g. CPU, GPU, RAM) during this period.

On the other hand, measuring such latency and energy consumption for generative AI models deployed on edge devices presents more challenges, due to the resource constraints and limited Software Development Kits (SDK). For instance, Google's MediaPipe Solution, while adept at cross-platform deployment, currently supports a limited subset of large language models. For Samsung Galaxy S24, it is challenging to deploy models with more than 2 billion parameters and run inference on NPUs through MediaPipe. Qualcomm's toolchain, though supporting Llama-2 inference with 7 billion parameters on NPUs, focuses more on devices with recent Qualcomm chipsets at the current stage. Thus, in the main experiment, we select Qualcomm's toolchain to evaluate energy consumption and latency using AI-optimized NPUs, as detailed in Section 4.1. Cross-chipset comparisons across various LLMs are presented in our full technical report[10].

3.3 Generative AI Models

We choose several popular generative AI models with various sizes and model architectures. These models are deployable both on cloud servers and edge devices. We begin with TrOCR, a light-weight transformer-based optical character recognition (OCR) model for reference, which generates character sequence from the input image. Next, we experiment with the diffusion models, a family of text-to-image generative AI models. More specifically, we consider the following four models, Stable Diffusion V2.1, Stable Diffusion V1.5, Riffusion , ControlNet, each with distinct architectural designs, application focuses, and training techniques. For text-to-text generative models, we deploy and profile the latency of Llama-2-7B, one of the established open-source large language models, both on-cloud and on-device . We choose the 7B version as it achieves a good tradeoff between memory requirements and model performance.

When deploying generative AI models in the cloud, further quantization is typically avoided to preserve optimal model performance, especially for models with small and medium sizes [12, 13]. However, the limited resources of edge devices (e.g. RAM), often necessitate additional modifications to adapt these models to hardware constraints. For example, the Qualcomm AI Hub[14] employs a hybrid 4-bit and 8-bit quantization approach for the Llama-2-7B model. The quantized model is optimized for Qualcomm chipset neural processing units (NPUs), and employs sequential loading of sharded models to address memory constraints. These optimizations, while designed to suit specific SoCs, do not always guarantee reduced latency or energy consumption on the AI servers hosted on cloud. Therefore, we evaluate the energy consumption and environmental footprint of state-of-the-practice representative deployment strategies across both cloud and edge platforms.

3.4 Dataset

Our experiment encompasses three distinct types of tasks: text generation, image generation, and optical character recognition. For each task, we select one widely used dataset for the inference of generative models. For the text generation task, we use the `vicgalle/alpaca-gpt4` dataset from the Hugging Face Hub. This dataset contains 52,000 entries, each with four columns. We combine the `instruction` and `input` columns to prepare the prompts. For the image generation task, we utilize the `poloclub/diffusiondb-Text` dataset. This dataset has several subsets, so we select one subset with 1,000 entries, employing the `prompt` column as input for inference. Finally, for the task optical character recognition (OCR), we use the `corto-ai/handwritten-text` dataset. This dataset contains 2,920 entries, each with handwritten text images in one column, which we use as input

for inference.

For the inference of each task, we sample 50 prompts from the corresponding dataset. To ensure the consistency in performance comparison, the same set of samples is used as input for the generative model hosted both on-cloud and on-device. Additionally, the sampled prompts or images are processed without batching operations for a fair comparison between the cloud and edge devices.

3.5 Latency Profiling Methodology

In addition to the model architecture, some hyperparameters of these generative models also play a key role in the latency measurement. For instance, the number of diffusion steps directly impacts the trade-off between the overall latency and generation quality. To ensure reproducibility, we explicitly state key hyperparameters: diffusion steps were consistently set to 50 for all models across cloud and edge deployments. The maximum token length for all LLMs was 1024. Our empirical results for Llama-2-7B, with an average token length of approximately 373, confirm the validity of this setting. With the latency estimates detailed as below, the energy consumption of each edge device is derived by multiplying its measured average power (e.g. 6.22 Watt for Samsung S24).

Profiling such end-to-end latency for generative models deployed on cloud is straightforward using standard tools. In contrast, profiling NPU-deployed models presents a challenge due to the lack of reliable software. We address this by profiling the latency of each component separately on Qualcomm AI Hub[14], then obtain the total latency according to the usage of each component. In this section, we delve into the detailed composition of the chosen generative AI models and how to profile the individual components.

3.5.1 Llama-2-7B

The Llama-2-7B model processed by Qualcomm is quantized and sharded into several components, to facilitate the deployment in edge devices. More concretely, suppose the prompt and token generator of Llama model are divided to N_p and N_g parts, we denote the i-th prompt generator and j-th token generator as p_i and g_j, respectively. The profiled latencies of sharded model p_i and g_j, obtained from Qualcomm AI Hub, are T_{p_i} and T_{g_j} respectively. To sum up, the overall latency is given by

$$T = \sum_{i=1}^{N_p} T_{p_i} + (N_t - 1) \cdot \sum_{j=1}^{N_g} T_{g_j} \tag{2}$$

where N_t is the number of tokens generated by Llama-2 model, depending on the input prompt. As the token length differs with different prompt inputs, the actual latency of Llama-2 also shows a significant variance. In our experiment, the number of generated tokens was determined by deploying the same model on GPUs with identical input prompts. Both the prompt and token generators were sharded into four parts ($N_p = 4$, $N_g = 4$).

3.5.2 Latent Diffusion Models

The selected diffusion models introduced in Section share a similar model architecture, composed of text encoder, the single-step diffusion model (e.g. U-Net or its variants) and the variational autoencoder (VAE). For simplicity, all their overall latency are denoted as below

$$T = T_{\text{TE}} + T_{\text{VAE}} + N_{df} \cdot T_{\text{U-Net}} \tag{3}$$

where N_{df} is the number of diffusion steps, $T_{\text{U-Net}}$ represents the single-step diffusion time, T_{TE} denotes the text encoder's inference time, and T_{VAE} denotes the inference time of VAE.

3.5.3 ControlNet

Similar to diffusion models, ControlNet employs a shared encoder-diffusion-decoder architecture. However, the distinguishing feature of ControlNet is its augmented diffusion setup, incorporating an additional module, denoted as C, to process contextual inputs. Consequently, the overall latency of the ControlNet module is given by

$$T = T_{\text{TE}} + T_{\text{VAE}} + N_{df} \cdot (T_{\text{C}} + T_{\text{U-Net}}) \tag{4}$$

where T_C denotes the inference time of module C, and the other parts are kept the same as the original diffusion model.

3.5.4 Text Recognition Model

The transformer based model for optical character recognition, TrOCR, is split into two components for on-device deployment. Specifically, the encoder and decoder of the transformer were loaded sequentially during inference. As there's no need to iterate over these components, the overall latency can be calculated by

$$T = T_{\text{En}} + T_{\text{De}} \tag{5}$$

where T_{En} and T_{De} denote the latency of the transformer's encoder and decoder, respectively.

4. EMPIRICAL RESULTS

In this section, we provide a summary of our results, suggesting that edge inference can potentially reduce the energy consumption and environmental footprints compared to the cloud counterpart.

4.1 Inference Time and Energy

For the generative models deployed on Google Colab, we measure their latency and energy consumption in an end-to-end measure using CodeCarbon [11]. Then we augment the measured energy with PUE $\delta = 1.10$ to take non-IT equipment energy into consideration. Such PUE factor matches the fleet-wide average annual PUE in Google, according to its 2024 ESG report [15]. For edge devices, we measure the end-to-end latency by comparing the difference between the system timestamp before and after executing generative models on CPUs or GPUs. In our experiment, we first fully charge the phone battery, then execute generative model inferences, and finally measure the energy consumed to recharge the battery back to its original state. By accounting for the charging efficiency, the overall energy consumption of generative models on edge devices considers the energy overhead for the charger's AC-DC conversions. According to the Level-VI efficiency regulation in the U.S. and the self-reporting charging efficiency by Samsung, we choose the charging efficiency of a wired charger as 0.8. On the NPUs, such end-to-end energy and latency measurement is technically challenging due to the lack of open-source solutions to readily deploy large language models on NPUs. Instead, we profile the latency of each component of generative models using Qualcomm AI hub, then derive the overall latency according to the usage of each component, as detailed in Section 3.5. The results

Model Name	NVIDIA L4		NVIDIA A100		Samsung S24 NPU		
	Latency (s)	Energy (J)	Latency (s)	Energy (J)	Latency (s)	Energy (J)	Energy Saving
TrOCR	0.03	3.24	0.02	3.39	0.11	**0.84**	75%
Stable Diffusion 1.5	3.35	539.36	2.24	714.86	4.64	**34.27**	95%
Stable Diffusion 2.1	6.95	1118.12	2.27	1154.89	9.57	**70.73**	94%
Riffusion	3.38	542.60	2.27	719.95	4.70	**34.72**	95%
ControlNet	6.64	1067.77	3.52	1292.34	14.03	**103.62**	92%
Llama-2-7B	23.56	3477.18	22.00	3803.71	35.50	**262.26**	93%

Table 1: Average inference time (seconds) and energy consumption (joules) for profiled AI models on three devices: Nvidia A100, Nvidia L4, and Samsung S24. The average token length of Llama-2 responses is **373**. The minimum energy values for each generative model are highlighted in bold. Energy savings of the Samsung S24 are calculated relative to the energy consumption of the Nvidia A100 when running the same model inputs.

Model Name	Total Carbon Footprint (Operational, Embodied)						
	NVIDIA L4		NVIDIA A100		Samsung S24		Carbon Saving
TrOCR	0.45	(0.35, 0.10)	0.45	(0.37, 0.08)	**0.24**	(0.09, 0.15)	46%
Stable Diffusion 1.5	69.44	(58.43, 11.01)	84.81	(77.44, 7.37)	**9.86**	(3.71, 6.15)	88%
Stable Diffusion 2.1	143.95	(121.13, 22.82)	132.56	(125.11, 7.45)	**20.35**	(7.66, 12.69)	85%
Riffusion	69.87	(58.78, 11.09)	85.44	(77.99, 7.45)	**9.99**	(3.76, 6.23)	88%
ControlNet	137.47	(115.67, 21.79)	151.57	(140.00, 11.57)	**29.81**	(11.23, 18.59)	80%
Llama-2-7B	454.04	(376.69, 77.34)	484.29	(412.07, 72.22)	**75.45**	(28.41, 47.04)	84%

Table 2: The average carbon footprint (gCO_2) per 1000 queries of the generative models, estimated with our model profiling results on Nvidia L4, Nvidia A100 and Samsung S24. The average carbon saving of Samsung S24 is calculated relative to the total carbon footprint of Nvidia A100 under the identical model inputs.

Model Name	Total Water Footprint (Scope 1, Scope 2)						
	NVIDIA L4		NVIDIA A100		Samsung S24		Water Saving
TrOCR	0.004	(0.001, 0.003)	0.004	(0.001, 0.003)	**0.001**	(0.000, 0.001)	81%
Stable Diffusion 1.5	0.621	(0.150, 0.471)	0.822	(0.199, 0.624)	**0.030**	(0.000, 0.030)	96%
Stable Diffusion 2.1	1.286	(0.311, 0.976)	1.329	(0.321, 1.008)	**0.062**	(0.000, 0.062)	95%
Riffusion	0.624	(0.151, 0.474)	0.828	(0.200, 0.628)	**0.030**	(0.000, 0.030)	96%
ControlNet	1.228	(0.297, 0.932)	1.487	(0.359, 1.128)	**0.090**	(0.000, 0.090)	94%
Llama-2-7B	4.001	(0.966, 3.035)	4.376	(1.057, 3.320)	**0.229**	(0.000, 0.229)	95%

Table 3: The average water footprint (L) per 1000 queries of the generative models, estimated with our model profiling results on Nvidia L4, Nvidia A100 and Samsung S24. The average water saving is calculated relative to the total water footprint of Nvidia A100 under the identical model inputs.

are summarized as Table 1, where we observe a significant energy saving when shifting the generative AI workload to the edge.

4.2 Environmental Footprints

In our experiment, we focus on two key environmental footprints — carbon footprint and water footprint — estimated based on the profiling results in Section 4.1.

Carbon Footprint: We choose the carbon intensity $\gamma_c = 390$ gCO_2/kWh for the operational carbon footprint, according to the average carbon emission by electricity generation in the U.S. [16]. For the two types of GPU servers we consider on Google Colab, the embodied carbon per GPU is set as 518 $kgCO_2$, on a par with [17]. We do not distinguish the embodied carbon difference between different GPUs, as the commonly-used embodied carbon footprints of GPUs are only rough estimates due to the lack of accurate data sources [18]. During the manufacturing process, distribution, and disposal, the carbon emission of Samsung S24 is estimated to be 39.2 $kgCO_2$ based on the Life-Cycle-Assessment result from Samsung's environmental report [19]. All these embodied carbon footprints are amortized over 5 years and the utilization ratios of cloud and mobile phone are set as 1 and 0.19, reflecting an estimated of 279 minutes per day for non-voice phone activities in the U.S. based on [20]. The higher utilization ratio in cloud servers offers some advantages for the cloud, as this can reduce the amortized carbon and incur a lower embodied carbon under the same condition. As shown in Table 2, the generative models deployed on our mobile phone shows significant reduction in both operational and embodied carbon emissions.

Water Footprint: Adopting the same methodology as the carbon footprint estimation, the water footprint is derived from energy consumption using water intensity γ_w. In our experiment, we mainly evaluate the scope-1 and scope-2 water consumption associated with deploying the generative AI model. Scope 1 consumption, primarily driven by the cooling requirements of cloud servers hosted in data centers, represents the direct water usage. Based on the data center water usage efficiency of Google [15] and Equinix (a major colocation provider) [7], we set the on-site water intensity factor as 1 L/kWh. As the edge device does not directly consume water, its scope-1 water consumption is set as zero. Moreover, the scope-2 water footprint is primarily attributed to the electricity generation process, with cloud servers and edge devices sharing the same water intensity. As reported in [21], the average grid water consumption factor in the U.S. is 3.142 L/kWh. Table 3 summarizes the scope-1 and -2 water consumptions of our selected generative AI models on cloud servers and edge devices.

Additional results that compare cloud inference and GPU-based edge inference are available in our full technical report [10], suggesting a similar trend that edge devices can be a competitive alternative to clouds in terms of the energy and

environmental footprints for generative AI inference.

5. CONCLUSION

This paper presents a case study to investigate the energy consumption and environmental footprints of selected generative AI models deployed in cloud and edge platforms. Our analysis focuses on the inference stage, where we profile the energy consumption and estimate the environmental footprint through detailed system modeling and publicly available data. Our findings suggest that potential energy saving and environmental benefits could be realized by shifting generative AI inference workloads from the cloud to edge devices. It is important to note that the generative AI models used in our experiment can be further optimized in both cloud and edge environments. Thus, the actual energy saving and environmental benefits may vary based on the specific optimization strategies implemented, and further research is needed.

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